Vishwa Karan Reddy Belaganti

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Computer Science Honors student focused on backend, full-stack, and applied AI/ML systems, with experience building datacenter management tools and a published Chrome Web Store extension.

EDUCATION

Texas Tech University, Honors College **Expected May 2027**
Bachelor of Science in Computer Science, Minor in Mathematics *GPA: 3.96/4.0*
Honors: President's Honors List (4 semesters), Dean's List, Terry G Myers Scholarship

PROFESSIONAL EXPERIENCE

Automation Intern, CtrlS Datacenters **Jun 2026 – Aug 2026**
Tier IV datacenter operator with 19 sites across India | Next.js, Three.js, Total.js, FastAPI, PostgreSQL, MongoDB

- Built a 3D CMDB with live data modeling every server and rack across all 19 datacenter sites (13,000+ VMs), replacing manual drill-down through tables and AG Grid; used by multiple teams.
- Created a floor-design editor that maps placed objects to database records, enabling 1:1 floor maps of existing infrastructure across all India locations for the first time; rollout has started.
- Contributed to the multi-tenant RBAC design in PostgreSQL and MongoDB that isolates client and administrative access.
- Gathered requirements firsthand, designed the Material Management System in Figma, and built a React prototype.

Grading Assistant (Data Structures), Texas Tech University **Sep 2025 – Jan 2026**

- Evaluated 700+ assignments for ~180 students, assessing correctness, runtime efficiency, and edge-case handling.

TECHNICAL PROJECTS

Chronos – Memory & Goal Tracking for AI Chats | *TypeScript, React, IndexedDB, Transformers.js*

- Built a Chrome extension for ChatGPT and Gemini that organizes conversations into topics, saves restorable checkpoints, and tracks goals across chats.
- Built an on-device topic-shift classifier on EmbeddingGemma embeddings, correctly classifying 82 of 99 test cases versus 29 for a naive baseline.
- Backed by 3,100+ automated tests and live ChatGPT/Gemini verification.

AgentOps – AI Agent Observability & Evaluation Platform | *Python, FastAPI, React, PostgreSQL, Redis*

- Built a multi-tenant platform with a Python SDK and a Redis Streams ingestion pipeline that reconstructs traces and computes cost and latency percentiles (p50–p99).
- Implemented an evaluation engine (rule-based checks plus LLM-as-a-judge), automated failure and root-cause detection, and safety-guarded replay for regression testing.
- Packaged with Docker Compose (API, workers, PostgreSQL, Redis) and Alembic migrations, with automated tests covering tenancy and security.

In-Memory PDF & Markdown Dropper – Published on Chrome Web Store | *Chrome MV3, pdf-lib, PDF.js*
chromewebstore.google.com/detail/okhaamedeamkdbgkcpfjohncoicneehd

- Shipped a Chrome extension that captures webpages and full ChatGPT, Gemini, Claude, and Perplexity conversations as vector PDFs or Markdown in memory, then drags them into any web drop area, like AI chats, without saving to disk.
- Created an in-browser PDF editor (highlight, redact, pen, page rotate/delete, searchable notes) with PDF merging.
- Added multi-tab batch capture; processing is fully client-side with zero user data collection.

Reinforcement Learning – PPO Extensions | *Python, PyTorch, Gymnasium, NumPy*

- Implemented PPO with critic ensembles and GAE, identifying 3 critics as the most stable setup on the Pendulum task.
- Added parallel data collection, cutting average training time from 291 s to 176 s (~40%), and a GRU recurrent actor that learned from partial observations, improving return from about -1700 to -1250.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C#, HTML/CSS
Frameworks: React, Next.js, Node.js, Express, FastAPI, PyTorch, Three.js, Tailwind CSS
Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase/Firestore, IndexedDB
Tools: Git, GitHub Actions, Docker, Linux, Playwright, Chrome Extensions, DevTools Protocol, Gemini API, Transformers.js